

Question	Answer	Marks
3(a)	force per unit mass	B1
3(b)(i)	$F = GMm / x^2$	C1
	$g = F / m$	A1
	$g = [GMm / x^2] / m = GM / x^2$ and G = gravitational constant	
3(b)(ii)	arrow drawn at P pointing directly towards the point mass	B1
3(b)(iii)	fields are in opposite directions	B1
	field strength at Q is four times the field strength at P	B1
3(c)	line starting at $(R, -g_0)$ and ending at $(L - R, +g_0)$	B1
	line passing through $(L / 2, 0)$	B1
	curve becoming shallower from R to $(L / 2)$ and then steeper from $(L / 2)$ to $(L - R)$	B1

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